TEN-FRAMES & PLACE VALUE ARITHMETIC LAB

5-Anchor Subitizing, Regrouping & Base-10 Manipulatives

CARD 01 / 10 • BASE-10

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

5-ANCHOR SUBITIZING & BRIDGING TO 10

Ten-frames visually organize numbers into 5-anchors:
Number = 5 + n and 10 = n + (10 − n)
Seeing 8 as 5 on top and 3 below anchors addition, turning counting into instant visual recognition!

CALIBRATED 14.00 MM SUBITIZING GRID

- Counter Diameter: 14.00 mm (2Ø) with two-tone face
- Stacking Datum: 1.85mm base plug & stacking socket
- Ten-Frame Grid: 2 × 5 array with 14.00 mm wells
- Base-10 Rods: 10Ø = 70.00 mm (ten 7mm units)

TWIN 10-FRAME STAGE LID & REGROUPING

Two side-by-side 10-frame stages model 2-digit sums and place-value regrouping across 10 on the lid!

5-ANCHOR SUBITIZING: 5 + 3 = 8 (TEN'S COMP = 2)



INSTANT SUBITIZING
Top Row: 5 Blue (anchor)
Bottom Row: 3 Red (8 total)

2 Empty Slots:
10 − 8 = 2 Ten's Complement
Bridge to 10: 8 + 5 = 13

2. KIT PARTS INVENTORY (34 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
10-Frame Trays	2x	140.0 × 56.0 mm (2×5 Grid)	Matte Charcoal
Two-Color Counters	20x	14.0 mm ø (Blue / Red)	Dual-Sided
Base-10 Rods (10s)	10x	70.0 × 7.00 mm (10U)	Warm Gold
Twin Stage Lid	1x	Side-by-Side 10-Frame	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 32 Manipulatives + Storage Bin + Twin Stage Lid = 34 Parts

DUAL-COLOR HIGH-CONTRAST PALETTE

- Blue Face: First addend or positive value.
- Red Face: Second addend or negative value.
- Warm Gold Rods: 10-unit place-value regrouping rods.
- High-Contrast Wells: Precision visual alignment.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Populate both 10-frames with 10 counters each (all 20 counters accounted for). Lay the 10 gold rods into the center trough. Any empty well instantly spots a piece. Twin stage lid snaps flush for silent zero-rattle carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

DECIMAL PLACE-VALUE REGROUPING THEOREM
Base-10 positional notation decomposes any integer into powers of 10 with digits 0 through 9.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 01 OF 10 • DOMAIN: PLACE VALUE & 5-ANCHOR SUBITIZING • 7.00mm 1U PITCH • 10-FRAME STACKING SLOTS

LESSON GUIDE: MAKE-A-TEN BRIDGE & 5-ANCHOR SUBITIZING

Step-by-Step Hands-On Tutorial: 8 + 5 = 13 via Bridging to 10

Grades K-3 Subitizing & Place Value • Hands-On Classroom & Home STEM Studio Series

CARD 01 / 10 • BASE-10

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/ten-frames/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.1.OA.C.6 (Make Ten)
- CCSS.MATH.CONTENT.2.NBT.B.5 (Place Value)
- TEKS 111.3.b.3.C (Compose/Decompose)

Grades K-3 Subitizing & Place Value

💡 **MATHEMATICAL INQUIRY**

How many empty spots are left in Frame A before you start moving counters from Frame B?

1 STEP 1: POPULATE 8 ON TEN-FRAME A

Place 8 blue counters in Ten-Frame A.
Fill the top row (5 blue) and bottom row (3 blue).
Instant Subitizing: 5 + 3 = 8 (zero counting needed!).
Notice: exactly 2 slots remain empty to complete the ten.

2 STEP 2: POPULATE 5 ON TEN-FRAME B

Place 5 red counters in Ten-Frame B (fills top row of 5).
We want to calculate 8 + 5 without counting by ones.
Frame A needs 2 counters to bridge to 10.

3-4 STEPS 3 & 4: BRIDGE TO 10 & REGROUP

3. BRIDGE TO 10: Move 2 red counters from B to A. Frame A is now 1 Ten (10)!
4. READ REMAINING ONES: Frame B has 3 red counters left. Total: 10 + 3 = 13!
Keystrokes: Ten-Frame Icon ([■] & [PV]) 8 [ENTER] 2 [+] 5 [+] [■] → 15.

5 CORE PRINCIPLE: ADDITION IS BRIDGING, NOT COUNTING

Counting on fingers by ones is slow and error-prone. The 10-frame proves that addition is decomposing numbers into friendly anchors: 8 + 5 = 8 + (2 + 3) = (8 + 2) + 3 = 10 + 3 = 13!
Once students see the 2 empty spaces in the 8-frame, they instantly know to borrow 2 from the 5!

★ EXTENSION INVESTIGATION: BRIDGE TO 10: 7 + 6

Put 7 in Frame A (3 empty). Put 6 in Frame B. Move 3 to make a ten. How many are left in B? 10 + 3 = 13! (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

BASE-10: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 34 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California